

Cross-Certification Strategy

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1. Purpose

Define the strategy for establishing cross-certification relationships with external CAs and PSC providers, enabling Goya Ledger certificates to be trusted by third parties and vice versa.

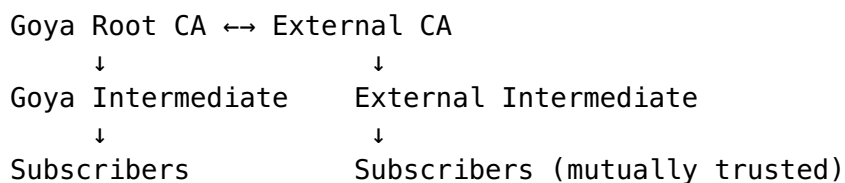
2. Context

Cross-certification is required for: - Interoperability with other Chilean PSC acreditados - International recognition (DS 261/2019 — mutual recognition with Argentina) - Integration with government PKI infrastructure (ClaveÚnica, Registro Civil) - Trust chain bridging between Goya's post-quantum algorithms and legacy CAs

3. Cross-Certification Models

3.1 Bilateral Cross-Certification

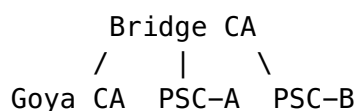
Two CAs issue cross-certificates for each other's root. Each CA trusts the other's subscribers.



Use case: Partnership with another Chilean PSC.

3.2 Bridge CA Model

A neutral bridge CA cross-certifies multiple CAs, creating a hub-and-spoke trust model.



Use case: Government-operated trust bridge for all Chilean PSC.

3.3 Trust List Inclusion

Goya’s root certificate is included in a trusted list maintained by a government or industry body.

Use case: Entidad Acreditadora TSL inclusion after accreditation.

4. Prerequisites for Cross-Certification

Requirement	Status	Notes
Accredited PSC status	Pending	Required before any cross-cert
CP/CPS published	Implemented	src/ pki_policy.rs, API at /policy/ cp
CRL in RFC 5280 format	Implemented	src/msp/ crl_rfc5280.rs
OCSP responder operational	Implemented	src/msp/ocsp.rs
CA hierarchy (root + intermediate)	Implemented	src/pki.rs CaHierarchy
TSL published	Implemented	src/tsl.rs, API at /tsl
Key ceremony completed	Framework ready	src/ pki_ceremony.rs
FIPS 140-3 validated module	Pending	Requires CMVP lab
RSA support for legacy interop	Not implemented	Ed25519/ML- DSA-65 only

5. Target Cross-Certification Partners

5.1 Priority 1 — Chilean PSC Acreditados

PSC	Services	Priority	Rationale
E-CERTCHILE	FEA, TSA, Biometría	High	Largest market share
ACEPTA.COM	FEA, TSA, Firma Móvil	High	Broadest service set
E-SIGN	FEA, TSA, Biometría	Medium	Post-quantum interest

5.2 Priority 2 — International

Entity	Country	Framework
Argentine CAs	Argentina	DS 261/2019 mutual recognition
EU Qualified TSPs	EU	eIDAS cross-recognition

5.3 Priority 3 — Government

Entity	Purpose
ClaveÚnica	Citizen identity verification integration
Registro Civil	RUT verification for RA identity proofing
SII	Tax authority document signing

6. Technical Requirements

6.1 Certificate Profile Alignment

- Cross-certificates must include `nameConstraints` to limit scope
- Policy mapping between Goya's CP OID and partner's CP OID
- `certificatePolicies` extension with both OIDs

6.2 Algorithm Interoperability

Scenario	Goya Algorithm	Partner Algorithm	Bridge
Legacy interop	Ed25519	RSA-2048/4096	Dual-signed cross-cert
PQ migration	ML-DSA-65	Ed25519/RSA	Hybrid certificate
PQ-native	ML-DSA-65	ML-DSA-65	Direct cross-cert

6.3 Revocation Interoperability

- Both parties must honor each other's CRL
- OSCP cross-queries for real-time status
- AIA (Authority Information Access) extensions pointing to partner's OSCP

7. Process for Establishing Cross-Certification

1. **Initiation:** Formal request between CAs with CP/CPS exchange
2. **Policy Review:** Compare CP/CPS for compatibility; identify policy mappings
3. **Technical Assessment:** Verify interoperability (algorithms, formats, revocation)
4. **Legal Agreement:** Cross-certification agreement covering liability, audit rights, termination
5. **Key Exchange:** Secure exchange of root certificates (in-person or witnessed ceremony)

6. **Cross-Certificate Issuance:** Each CA issues a cross-certificate for the other's root
7. **Publication:** Update TSL, CRL distribution points, and AIA extensions
8. **Testing:** End-to-end certificate chain validation from both sides
9. **Monitoring:** Ongoing compliance monitoring and annual review

8. Termination Procedures

1. 90-day advance written notice to partner
2. Revoke cross-certificate
3. Publish updated CRL within 1 hour
4. Update TSL to remove partner
5. Notify affected subscribers
6. Notify Entidad Acreditadora
7. Archive all cross-certification records (7-year retention)

9. Regulatory References

- Ley 19.799 Art. 14 — Mutual recognition of electronic signatures
- DS 261/2019 — Mutual recognition agreement with Argentina
- ETSI TS 102 042 §7.1.3 — Cross-certification policy
- RFC 4158 — Internet X.509 PKI: Certification Path Building
- RFC 5280 §4.2.1.4 — Certificate Policies extension

10. Document Control

Version	Date	Author	Changes
1.0	2026-08-05	PKI Administrator	Initial draft